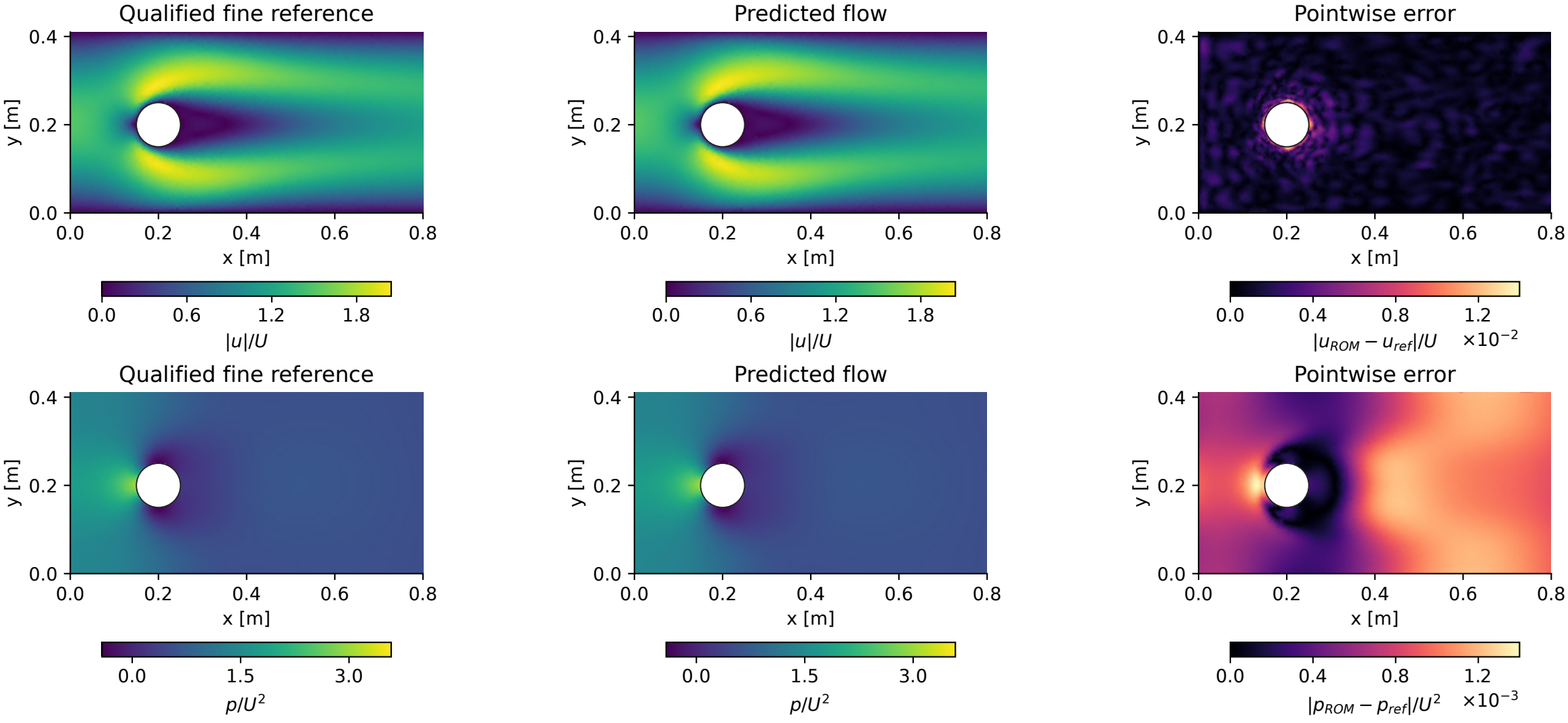

Refined neural-bank ROM: flow-solve output — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Predicted flow
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
0.0
0.6
1.2
1.8
|u|/U
|u_ROM − u_ref|/U
×10^−2
Qualified fine reference
Predicted flow
Pointwise error
0.0
1.5
3.0
p/U^2
|p_ROM − p_ref|/U^2
×10^−3
Longer-trained neural velocity bank plus FE pressure/stability spaces. Divergence remains above target.
Display interpolation; error scales differ. Errors against stored reference do not certify equal continuum accuracy. Development cases.